



2018 ANNUAL EXAMINATION CHEMISTRY Form V

STRUCTURE OF PAPER

- Section I: Multiple Choice (12 marks)
Answer all questions on the Multiple-choice Answer Sheet.
- Section II: Extended Response (72 marks)
Answer all parts of the questions in the spaces provided in the Examination Paper.
- Board approved calculators may be used

EXAMINATION

Date: Friday 7th September
Duration: 2 hours
Marks: 84

CHECKLIST

Each boy should have the following:

- ☐ 1 Examination Paper
- ☐ 1 Multiple Choice Answer Sheet
- ☐ 1 Data and Formula Sheet

EXAM INSTRUCTIONS

- Remove the centre staple and hand in all parts of the paper, in order, in one bundle.
- WRITE YOUR **NAME** AND **CLASS NUMBER** IN THE SPACE PROVIDED AT THE TOP OF EACH SEPARATE SECTION.

CLASS NUMBER	1	2	3	4	5	6	7	8
Class	5CY201	5CY202	5CY203	5CY204	5CY205	5CY206	5CY207	5CY208
Master Initials	EJS	CXS	CRMR	ZI/MTK	TW	AKBB	BED	DBD

Examiners: TW, AKBB, BED, CXS.

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SECTION I: MULTIPLE CHOICE

Attempt ALL Questions
Use the Multiple-Choice Answer Sheet.

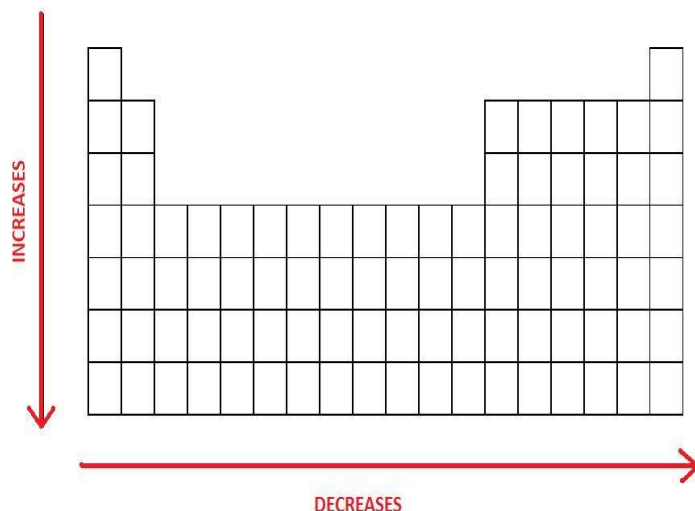
- 1 Which of the following substances contains atoms bonded together by covalent bonds only?
 - (A) MgO
 - (B) SiO_2
 - (C) NH_4NO_3
 - (D) NaCl

- 2 Which of the following pairs of electron configurations correspond to a pair of atoms that would be expected to form a covalent bond?
 - (A) $1s^2 2s^2 2p^5$ and $1s^2 2s^2 2p^4$
 - (B) $1s^2 2s^2 2p^6 3s^2 3p^6$ and $1s^2$
 - (C) $1s^2 2s^2 2p^6 3s^1$ and $1s^2 2s^2 2p^6 3s^2 3p^5$
 - (D) $1s^2 2s^2 2p^6 3s^2 3p^3$ and $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$

- 3 Which of the following substances would be **least** polar?
 - (A) CCl_4
 - (B) CFCl_3
 - (C) HCl
 - (D) Cl_2O

- 4 A 450 mL sample of 1.50 M sodium acetate solution was evaporated to achieve a final concentration of 2.25 M. What is the final volume of the solution?
 - (A) 150 mL
 - (B) 200 mL
 - (C) 300 mL
 - (D) 400 mL

- 5 The diagram shows a change in a property of atoms down a group and across a period of the Periodic Table.

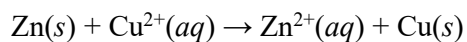


The property whose change is shown in the diagram could be

- (A) first ionisation energy.
 - (B) relative atomic mass.
 - (C) electronegativity.
 - (D) atomic radius.
- 6 Which of the following correctly identifies a process that occurs at an electrode?
- (A) Reduction at the anode.
 - (B) Electron loss at the cathode.
 - (C) Electron gain at the anode.
 - (D) Oxidation at the anode.

- 7 In which of the following situations is the manganese-containing species oxidised?
- (A) $2\text{MnO}_4^-(aq) + 16\text{H}^+(aq) + 10\text{Cl}^-(aq) \rightarrow 2\text{Mn}^{2+}(aq) + 8\text{H}_2\text{O}(l) + 5\text{Cl}_2(g)$
- (B) $2\text{Mn}^{2+}(aq) + 4\text{OH}^-(aq) + \text{O}_2(g) \rightarrow 2\text{MnO}_2(s) + 2\text{H}_2\text{O}(l)$
- (C) $\text{MnO}_4^{2-}(aq) + \text{H}_2(g) \rightarrow \text{MnO}_2(s) + 2\text{OH}^-(aq)$
- (D) $\text{MnO}_2(s) + 2\text{C}(s) \rightarrow 2\text{CO}(g) + \text{Mn}(s)$
- 8 A balloon filled with 1.00 L of air at 29.0 °C was cooled to 15.0 °C. Assuming that the pressure remains constant, what is the final volume of the balloon?
- (A) 0.349 L
- (B) 0.517 L
- (C) 0.954 L
- (D) 1.049 L
- 9 How many nitrogen atoms are in 1.34 mol of copper(II) nitrate?
- (A) 2.25×10^{23} atoms
- (B) 4.49×10^{23} atoms
- (C) 8.07×10^{23} atoms
- (D) 1.61×10^{24} atoms
- 10 Nitrogen and hydrogen gas react to form ammonia (NH_3). If 3.0 L of nitrogen is added to excess hydrogen at 100 kPa and 293 K, what volume of ammonia will be produced, assuming that all of the nitrogen is consumed?
- (A) 1.5 L
- (B) 3.0 L
- (C) 6.0 L
- (D) 9.0 L

- 11 When 9.86 J of energy is transferred as heat to a sample of aluminium metal, the temperature of the sample increases from 23.2 °C to 30.5 °C. The specific heat capacity of aluminium is 0.90 J K⁻¹ g⁻¹. What is the mass of the aluminium metal sample?
- (A) 0.67 g
(B) 0.82 g
(C) 1.2 g
(D) 1.5 g
- 12 Given that the standard enthalpy of formation for Cu²⁺(aq) is +64.4 kJ mol⁻¹ and that of Zn²⁺(aq) is -152.4 kJ mol⁻¹, what is the standard enthalpy of reaction for the following reaction?



- (A) -216.8 kJ mol⁻¹
(B) -88.0 kJ mol⁻¹
(C) +88.0 kJ mol⁻¹
(D) +216.8 kJ mol⁻¹

SECTION II: Attempt ALL Questions

Name:

Class:

Answer the questions in the spaces provided.
Show all relevant working in questions involving calculations.

Question 13 (4 marks)

Marks

The radioisotope, potassium-40 decays to form a beta particle and calcium-40.

- (a) Write a balanced nuclear equation for this reaction.

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- (b) Contrast two features of a beta particle with those of an alpha particle.

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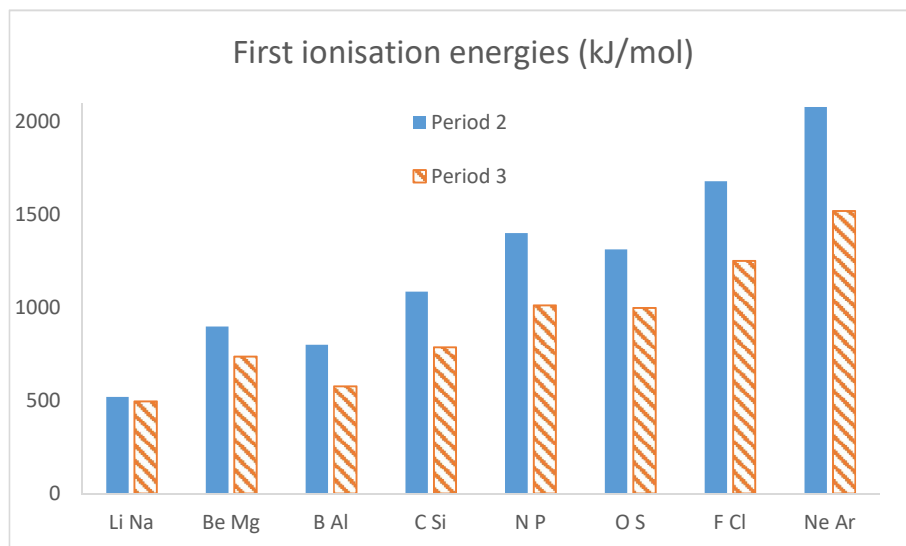
- (c) Give the electronic structure of an atom of potassium in spdf orbital notation.

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Question 14 (5 marks)**Marks**

The table below shows the first ionisation energies of the Period 2 and Period 3 elements, measured in kJ mol^{-1} .



- (a) Explain the general trend in first ionisation energy across Period 2.

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- (b) Explain why the elements of Period 3 have smaller first ionisation energies than the corresponding elements of Period 2.

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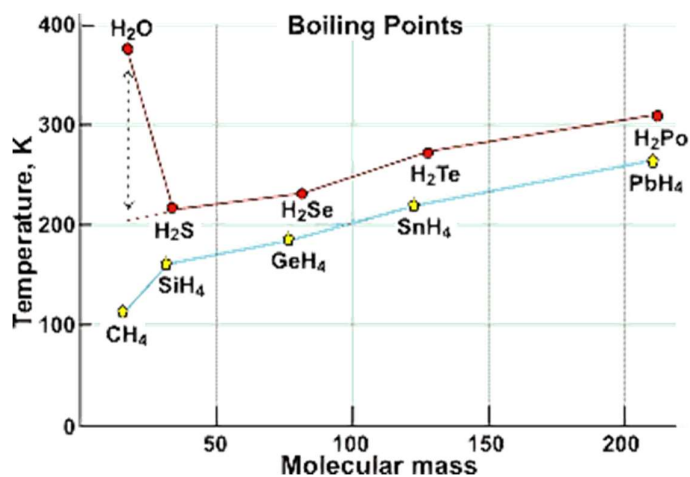
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Question 15 (7 marks)**Marks**

The graphs below show the relative boiling points of hydrides of elements in Groups 4 and 6 of the Periodic Table.



- (a) Draw a Lewis structure or electron dot formula to represent a molecule of hydrogen sulfide, (H₂S).

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- (b) Explain why H₂S is polar whereas SiH₄ is not.

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Question continued on next page.

Question continued.**Marks**

- (c) Explain the trend in the boiling points of the Group 6 hydrides from H_2S to H_2Po .

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- (d) Explain why the boiling point of water is significantly higher than would be expected from the trend for the other Group 6 hydrides.

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SECTION II: Attempt ALL Questions

Name:

Class:

Answer the questions in the spaces provided.
Show all relevant working in questions involving calculations.

Question 16 (2 marks)

Marks

Briefly explain how a process employed by Aboriginal or Torres Strait Islander peoples detoxifies poisonous food items.

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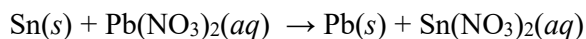
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Question 17 (3 marks)

Consider the following balanced chemical equation:



- (a) Does the reaction in the equation above proceed spontaneously? Justify your answer.

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- (b) Identify the spectator ion in the equation in part (a).

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- (c) Identify the reducing agent (reductant), for the equation in part (a).

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Question 18 (6 marks)**Marks**

A student conducts an investigation to determine the voltage of a galvanic cell made from half-cells containing magnesium and silver, and their respective ions.

- (a) Draw a labelled diagram of the experimental set-up. Indicate on your diagram the direction of electron flow and the direction of cation movement.

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- (b) Calculate the standard cell potential for this galvanic cell.

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- (c) The experimental voltage obtained from galvanic cells is often different to that indicated from standard reduction potentials. Suggest a reason why this may be.

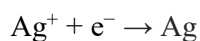
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Question 19 (5 marks)**Marks**

In a metal displacement reaction following half reactions were observed to occur:



- (a) Write a net ionic equation for this reaction, including all states.

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- (b) A strip of copper was placed into a solution containing silver nitrate. After the reaction was complete the silver was removed from the surface of the copper strip and the copper strip was removed. The resulting suspension, containing solid silver, was filtered.

copper strip before reaction	2.298 g
copper strip after reaction	2.216 g
filter paper before filtration	1.662 g
filter paper after filtration	2.002 g

- (i) Calculate the chemical amount (in mol) of silver produced and of copper consumed.

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- (ii) Calculate the experimental mole ratio of silver to copper and account for any deviation from ideal results.

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SECTION II: Attempt ALL Questions

Name:

Class:

Answer the questions in the spaces provided.
Show all relevant working in questions involving calculations.

Question 20 (4 marks)

Marks

A compound was found to contain 24.8% carbon, 2.0% hydrogen and 73.2% chlorine.

- (a) Determine the empirical formula of this compound.

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- (b) The molar mass of this compound is 96.9 g mol^{-1} . Determine the molecular formula of this compound.

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Question 21 (2 marks)

Chromium has four stable isotopes, whose relative isotopic masses and percentage abundances are shown in the table below.

Isotope	Isotopic mass (u)	Percentage abundance (%)
^{50}Cr	49.946	4.3450
^{52}Cr	51.941	83.789
^{53}Cr	52.941	9.5012
^{54}Cr	53.939	2.3657

Calculate the relative atomic mass of chromium.

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Question 22 (5 marks)**Marks**

In an experiment, 2.369 g of aluminium hydroxide reacts with 60.0 mL of 0.714 M sulfuric acid to produce a salt solution and water.

- (a) Write a balanced chemical equation for this reaction.

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- (b) Calculate the chemical amount (in mol) of the salt produced.

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4**Question 23** (4 marks)

A 33.34 g sample of a diatomic gas was stored in a 20.0 L vessel at 150 kPa and 30 °C. What is the identity of this gas?

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Question 24 (5 marks)**Marks**

Antacids contain calcium carbonate and are used to neutralise stomach acid. The mass of one antacid tablet was found to be 1.30 g. To determine the amount of calcium carbonate present, this tablet was crushed into a fine powder and excess hydrochloric acid was added to it. After the reaction was complete, there was a decrease in mass of 0.220 g.

- (a) Write a balanced equation for the reaction between calcium carbonate and hydrochloric acid.

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- (b) Calculate the mass (in g) of calcium carbonate present in the antacid tablet.

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- (c) Calculate the percentage by mass of calcium carbonate in the antacid tablet.

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- (d) At 100 kPa and 296 K, what volume of gas would have been produced?

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SECTION II: Attempt ALL Questions

Name:

Class:

Answer the questions in the spaces provided.
Show all relevant working in questions involving calculations.

Question 25 (4 marks)

Marks

	silicon tetrachloride	silicon dioxide
Melting point (°C)	−68	1713

Explain the melting points of silicon tetrachloride and silicon dioxide in terms of their structure and bonding.

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Question 26 (6 marks)**Marks**

5.000 g of silver nitrate is dissolved in 100.0 g of water, initially at 25.0 °C.

- (a) If the enthalpy of solution ($\Delta_{\text{sol}}H$) for silver nitrate is +22.6 kJ mol⁻¹, what is the final temperature of the solution?

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- (b) Given that silver nitrate is soluble in water at room temperature, what must be the sign of the entropy of solution ($\Delta_{\text{sol}}S$) for silver nitrate? Explain your answer.

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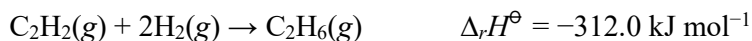
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Question 27 (2 marks)**Marks**

The reaction of acetylene (C_2H_2) with hydrogen (H_2) can produce either ethylene (C_2H_4) or ethane (C_2H_6):



Industrially, ethylene and hydrogen are produced from ethane on a massive scale.

What is $\Delta_r H$ for the reaction of ethane (C_2H_6) to form ethylene (C_2H_4) and H_2 ?

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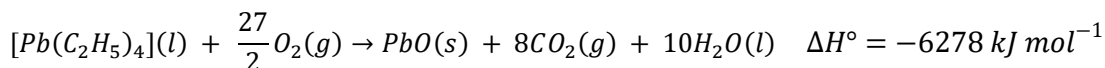
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2**Question 28** (3 marks)

Tetraethyllead $[\text{Pb}(\text{C}_2\text{H}_5)_4]$ was formally used as a fuel additive, but its use in automotive petrol has been banned in Australia since 2002. It combusts as follows:



	standard enthalpy of formation (kJ mol^{-1})
$\text{CO}_2(\text{g})$	-393.5
$\text{H}_2\text{O}(\text{l})$	-285.8
$[\text{Pb}(\text{C}_2\text{H}_5)_4](\text{l})$	+53.0

Given the information in the table above, calculate the standard enthalpy of formation of $\text{PbO}(\text{s})$.

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Question 29 (5 marks)**Marks**

Widgiemoolthalite is a rare blue-green mineral first identified in Widgiemooltha, Western Australia, in 1992. It contains three different ions, none of which contains atoms of more than two elements.

When 2.000 g of widgiemoolthalite reacts with excess hydrochloric acid it dissolves completely and releases 0.5354 g of carbon dioxide. Strongly heating 2.000 g of widgiemoolthalite produces water vapour, carbon dioxide and 1.1358 g of nickel(II) oxide as the only products.

Calculate the empirical formula for widgiemoolthalite, expressing your answer in terms of the ions and molecules present.

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END OF EXAMINATION

Formulae sheet, Data sheet and Periodic Table overleaf

Chemistry

FORMULAE SHEET

$$n = \frac{m}{MM}$$

$$q = mC\Delta T$$

$$pK_a = -\log_{10}[K_a]$$

$$c = \frac{n}{v}$$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$A = \epsilon lc = \log_{10} \frac{I_o}{I}$$

$$PV = nRT$$

$$\text{pH} = -\log_{10}[\text{H}^+]$$

Avogadro constant, N_A $6.022 \times 10^{23} \text{ mol}^{-1}$

Volume of 1 mole ideal gas: at 100 kPa and

at 0°C (273.15 K) 22.71 L

at 25°C (298.15 K) 24.79 L

Gas constant $8.314 \text{ J mol}^{-1} \text{ K}^{-1}$

Ionisation constant for water at 25°C (298.15 K), K_w 1.0×10^{-14}

Specific heat capacity of water $4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

DATA SHEET

Solubility constants at 25°C

Compound	K_{sp}	Compound	K_{sp}
Barium carbonate	2.58×10^{-9}	Lead(II) bromide	6.60×10^{-6}
Barium hydroxide	2.55×10^{-4}	Lead(II) chloride	1.70×10^{-5}
Barium phosphate	1.3×10^{-29}	Lead(II) iodide	9.8×10^{-9}
Barium sulfate	1.08×10^{-10}	Lead(II) carbonate	7.40×10^{-14}
Calcium carbonate	3.36×10^{-9}	Lead(II) hydroxide	1.43×10^{-15}
Calcium hydroxide	5.02×10^{-6}	Lead(II) phosphate	8.0×10^{-43}
Calcium phosphate	2.07×10^{-29}	Lead(II) sulfate	2.53×10^{-8}
Calcium sulfate	4.93×10^{-5}	Magnesium carbonate	6.82×10^{-6}
Copper(II) carbonate	1.4×10^{-10}	Magnesium hydroxide	5.61×10^{-12}
Copper(II) hydroxide	2.2×10^{-20}	Magnesium phosphate	1.04×10^{-24}
Copper(II) phosphate	1.40×10^{-37}	Silver bromide	5.35×10^{-13}
Iron(II) carbonate	3.13×10^{-11}	Silver chloride	1.77×10^{-10}
Iron(II) hydroxide	4.87×10^{-17}	Silver carbonate	8.46×10^{-12}
Iron(III) hydroxide	2.79×10^{-39}	Silver hydroxide	2.0×10^{-8}
Iron(III) phosphate	9.91×10^{-16}	Silver iodide	8.52×10^{-17}
		Silver phosphate	8.89×10^{-17}
		Silver sulfate	1.20×10^{-5}

Aylward and Findlay, *SI Chemical Data* (5th Edition) is the principal source of data for this examination paper. Some data may have been modified for examination purposes.

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{K}(s)$	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Ba}(s)$	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Ca}(s)$	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{Na}(s)$	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Mg}(s)$	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	$\text{Al}(s)$	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Mn}(s)$	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(g) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Zn}(s)$	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Fe}(s)$	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Ni}(s)$	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Sn}(s)$	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Pb}(s)$	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(g)$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(aq) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Cu}(s)$	0.34 V
$\frac{1}{2}\text{O}_2(g) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{Cu}(s)$	0.52 V
$\frac{1}{2}\text{I}_2(s) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(aq) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{Ag}(s)$	0.80 V
$\frac{1}{2}\text{Br}_2(l) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(aq) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(g) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(g) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(aq) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(g) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

PERIODIC TABLE OF THE ELEMENTS

1		KEY																2	
H 1.008 Hydrogen		Atomic Number Symbol Standard Atomic Weight Name																He 4.003 Helium	
3		79																9	
Li 6.941 Lithium		Au 197.0 Gold																F 19.00 Fluorine	
11		26																17	
Na 22.99 Sodium		Fe 55.85 Iron																Cl 35.45 Chlorine	
12		25																16	
Mg 24.31 Magnesium		Mn 54.94 Manganese																S 32.07 Sulfur	
19		24																35	
K 39.10 Potassium		Cr 52.00 Chromium																Br 79.90 Bromine	
20		23																34	
Ca 40.08 Calcium		V 50.94 Vanadium																Se 78.96 Selenium	
37		22																53	
Rb 85.47 Rubidium		Ti 47.87 Titanium																I 126.9 Iodine	
38		21																85	
Sr 87.61 Strontium		Sc 44.96 Scandium																At 210 Astatine	
55		20																86	
Cs 132.9 Caesium		Zr 91.22 Zirconium																Po 210 Polonium	
87		39																118	
Fr Francium		Y 88.91 Yttrium																Ts 289 Tennessine	
88		40																Lv 293 Livermorium	
Ra Radium		Nb 92.91 Niobium																Og 284 Oganesson	
89		41																294	
Ac Actinoids		Mo 95.96 Molybdenum																295	
90		Tc 98 Technetium																296	
91		Ru 101.1 Ruthenium																297	
92		Rh 102.9 Rhodium																298	
93		Pd 106.4 Palladium																299	
94		Ag 107.9 Silver																300	
95		Cd 112.4 Cadmium																301	
96		In 114.8 Indium																302	
97		Sn 118.7 Tin																303	
98		Sb 121.8 Antimony																304	
99		Te 127.6 Tellurium																305	
100		I 126.9 Iodine																306	
101		Xe 131.3 Xenon																307	
102		Kr 83.8 Krypton																308	
103		Br 79.9 Bromine																309	
104		Se 78.9 Selenium																310	
105		As 74.9 Arsenic																311	
106		Ge 72.6 Germanium																312	
107		Ga 69.7 Gallium																313	
108		Zn 65.4 Zinc																314	
109		Cu 63.5 Copper																315	
110		Ni 58.7 Nickel																316	
111		Co 58.9 Cobalt																317	
112		Fe 55.8 Iron																318	
113		Mn 54.9 Manganese																319	
114		Cr 52.0 Chromium																320	
115		V 50.9 Vanadium																321	
116		Ti 47.9 Titanium																322	
117		Sc 44.9 Scandium																323	
118		Ca 40.1 Calcium																324	
119		K 39.1 Potassium																325	
120		Ar 39.9 Argon																326	
121		Cl 35.5 Chlorine																327	
122		S 32.1 Sulfur																328	
123		P 31.0 Phosphorus																329	
124		Si 28.1 Silicon																330	
125		Al 26.9 Aluminum																331	
126		Mg 24.3 Magnesium																332	
127		Na 22.9 Sodium																333	
128		Ne 20.2 Neon																334	
129		F 19.0 Fluorine																335	
130		O 16.0 Oxygen																336	
131		N 14.0 Nitrogen																337	
132		C 12.0 Carbon																338	
133		B 10.8 Boron																339	
134		Be 9.0 Beryllium																340	
135		Li 6.9 Lithium																341	
136		H 1.0 Hydrogen																342	

Lanthanoids

57 La 138.9 Lanthanum	58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.1 Ytterbium	71 Lu 175.0 Lutetium
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Actinoids

89 Ac Actinium	90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np Neptunium	94 Pu Plutonium	95 Am Americium	96 Cm Curium	97 Bk Berkelium	98 Cf Californium	99 Es Einsteinium	100 Fm Fermium	101 Md Mendelevium	102 No Nobelium	103 Lr Lawrencium
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Standard atomic weights are abridged to four significant figures.

Elements with no reported values in the table have no stable nuclides.

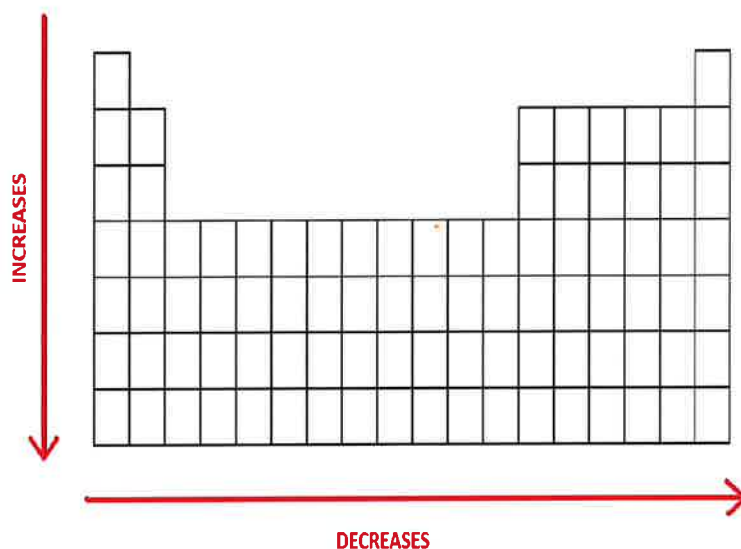
Information on elements with atomic numbers 113 and above is sourced from the International Union of Pure and Applied Chemistry Periodic Table of the Elements (November 2016 version). The International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of all other data. Some data may have been modified.

SECTION I: MULTIPLE CHOICE

Attempt ALL Questions
Use the Multiple-Choice Answer Sheet.

- 1 Which of the following substances contains atoms bonded together by covalent bonds only?
- (A) MgO
(B) SiO₂
(C) NH₄NO₃
(D) NaCl
- 2 Which of the following pairs of electron configurations correspond to a pair of atoms that would be expected to form a covalent bond?
- (A) $1s^2 2s^2 2p^5$ and $1s^2 2s^2 2p^4$
(B) $1s^2 2s^2 2p^6 3s^2 3p^6$ and $1s^2$
(C) $1s^2 2s^2 2p^6 3s^1$ and $1s^2 2s^2 2p^6 3s^2 3p^5$
(D) $1s^2 2s^2 2p^6 3s^2 3p^3$ and $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$
- 3 Which of the following substances would be **least** polar?
- (A) CCl₄
(B) CFCl₃
(C) HCl
(D) Cl₂O
- 4 A 450 mL sample of 1.50 M sodium acetate solution was evaporated to achieve a final concentration of 2.25 M. What is the final volume of the solution?
- (A) 150 mL
(B) 200 mL
(C) 300 mL
(D) 400 mL

- 5 The diagram shows a change in a property of atoms down a group and across a period of the Periodic Table.

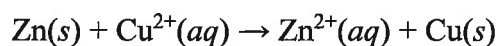


The property whose change is shown in the diagram could be

- (A) first ionisation energy.
 - (B) relative atomic mass.
 - (C) electronegativity.
 - (D) atomic radius.
- 6 Which of the following correctly identifies a process that occurs at an electrode?
- (A) Reduction at the anode.
 - (B) Electron loss at the cathode.
 - (C) Electron gain at the anode.
 - (D) Oxidation at the anode.

- 7 In which of the following situations is the manganese-containing species oxidised?
- (A) $2\text{MnO}_4^-(aq) + 16\text{H}^+(aq) + 10\text{Cl}^-(aq) \rightarrow 2\text{Mn}^{2+}(aq) + 8\text{H}_2\text{O}(l) + 5\text{Cl}_2(g)$
- (B) $2\text{Mn}^{2+}(aq) + 4\text{OH}^-(aq) + \text{O}_2(g) \rightarrow 2\text{MnO}_2(s) + 2\text{H}_2\text{O}(l)$
- (C) $\text{MnO}_4^{2-}(aq) + \text{H}_2(g) \rightarrow \text{MnO}_2(s) + 2\text{OH}^-(aq)$
- (D) $\text{MnO}_2(s) + 2\text{C}(s) \rightarrow 2\text{CO}(g) + \text{Mn}(s)$
- 8 A balloon filled with 1.00 L of air at 29.0 °C was cooled to 15.0 °C. Assuming that the pressure remains constant, what is the final volume of the balloon?
- (A) 0.349 L
- (B) 0.517 L
- (C) 0.954 L
- (D) 1.049 L
- 9 How many nitrogen atoms are in 1.34 mol of copper(II) nitrate?
- (A) 2.25×10^{23} atoms
- (B) 4.49×10^{23} atoms
- (C) 8.07×10^{23} atoms
- (D) 1.61×10^{24} atoms
- 10 Nitrogen and hydrogen gas react to form ammonia (NH_3). If 3.0 L of nitrogen is added to excess hydrogen at 100 kPa and 293 K, what volume of ammonia will be produced, assuming that all of the nitrogen is consumed?
- (A) 1.5 L
- (B) 3.0 L
- (C) 6.0 L
- (D) 9.0 L

- 11 When 9.86 J of energy is transferred as heat to a sample of aluminium metal, the temperature of the sample increases from 23.2 °C to 30.5 °C. The specific heat capacity of aluminium is 0.90 J K⁻¹ g⁻¹. What is the mass of the aluminium metal sample?
- (A) 0.67 g
(B) 0.82 g
(C) 1.2 g
(D) 1.5 g
- 12 Given that the standard enthalpy of formation for Cu²⁺(aq) is +64.4 kJ mol⁻¹ and that of Zn²⁺(aq) is -152.4 kJ mol⁻¹, what is the standard enthalpy of reaction for the following reaction?



- (A) -216.8 kJ mol⁻¹
(B) -88.0 kJ mol⁻¹
(C) +88.0 kJ mol⁻¹
(D) +216.8 kJ mol⁻¹

SECTION II: Attempt ALL Questions

CRIB

Name: _____

Class: _____

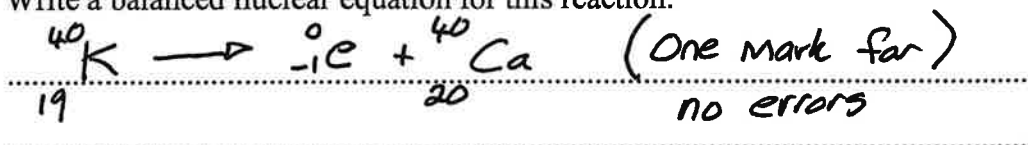
Answer the questions in the spaces provided.
Show all relevant working in questions involving calculations.

Question 13 (4 marks)

Marks

The radioisotope, potassium-40 decays to form a beta particle and calcium-40.

- (a) Write a balanced nuclear equation for this reaction.



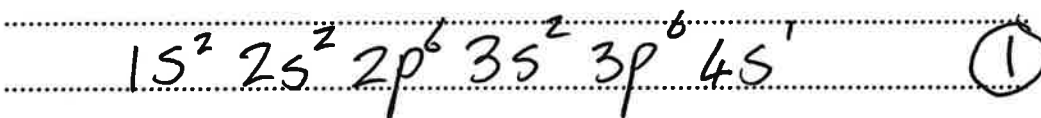
1

- (b) Contrast two features of a beta particle with those of an alpha particle.

Beta	-1 charge	0 mass	less ionising	more penetrative
Alpha	+2 charge	mass of 4	more ionising	less penetrative

(*) 1 mark for beta is ${}_{-1}^0\text{e}$ & alpha is ${}_{2}^4\text{He}$

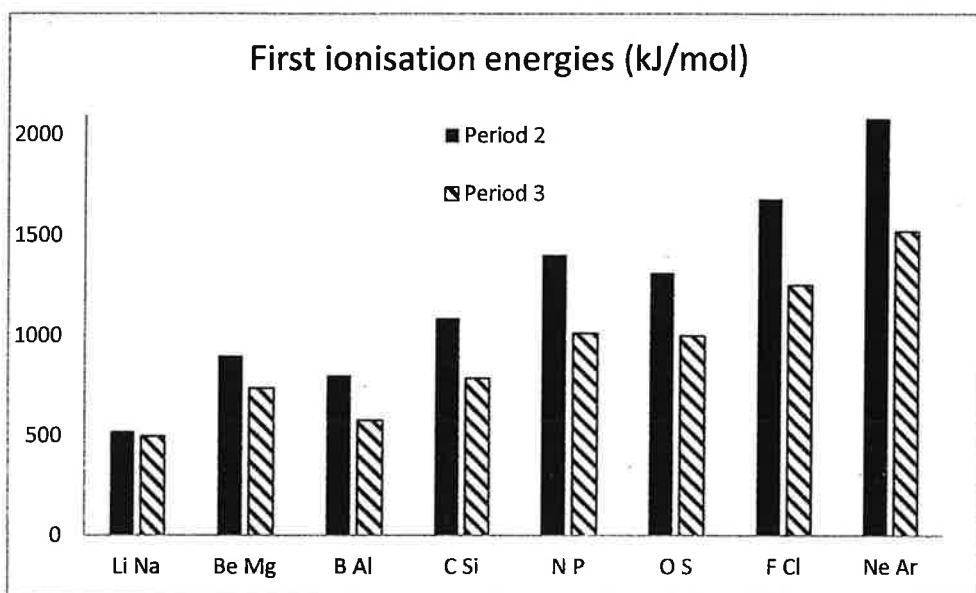
- (c) Give the electronic structure of an atom of potassium in spdf orbital notation.



1

Question 14 (5 marks)**Marks**

The table below shows the first ionisation energies of the Period 2 and Period 3 elements, measured in kJ mol^{-1} .



- (a) Explain the general trend in first ionisation energy across Period 2.

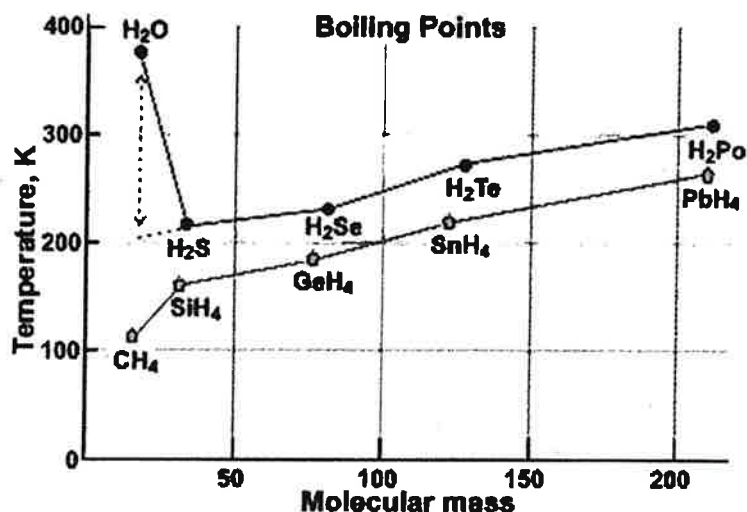
Increasing across period
 because : 1. Greater nuclear charge — ①
 2. Electrons removed from same shell ①
 3. ∴ More energy is required to remove electron ①

- (b) Explain why the elements of Period 3 have smaller first ionisation energies than the corresponding elements of Period 2.

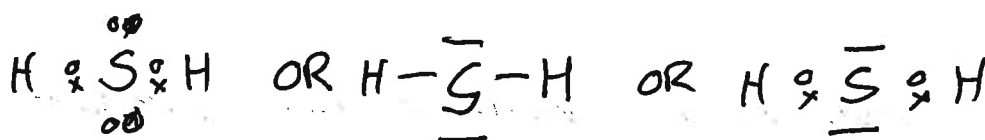
Period 3: valence electrons are further from the nucleus ①
 OR have an extra shell of electrons shielding
 ∴ the attractive nuclear force
 ∴ Less energy is required to remove one electron ①

Question 15 (7 marks)**Marks**

The graphs below show the relative boiling points of hydrides of elements in Groups 4 and 6 of the Periodic Table.



- (a) Draw a Lewis structure or electron dot formula to represent a molecule of hydrogen sulfide, (H₂S).



1

- (b) Explain why H₂S is polar whereas SiH₄ is not.

1 mark - H₂S is bent or not symmetrical

SiH₄ is tetrahedral or symmetrical

1 mark - uneven charge distribution
or words to that effect.

2

Question continued.

Marks

- (c) Explain the trend in the boiling points of the Group 6 hydrides from H_2S to H_2Po .

Increasing strength of dispersion forces ①

2

Owing to increasing number of electrons ①

*didn't give increased size (or mass)

(One mark if increasing boiling point is linked with intermolecular forces increasing)

(* If student identified dipole/dipole or H-bonds in addition to dispersion as playing a role then minus 1 mark)

- (d) Explain why the boiling point of water is significantly higher than would be expected from the trend for the other Group 6 hydrides.

Hydrogen bonds between H_2O molecules and these are the strongest intermolecular forces ①

Other Group 6 hydrides have weaker i.f. ①

SECTION II: Attempt ALL Questions

Name:

CR1B

Class:

MTK

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Question 16 (2 marks)

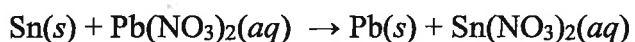
Marks

Briefly explain how a process employed by Aboriginal or Torres Strait Islander peoples detoxifies poisonous food items.

1 - process eg. soak in water (after grinding) for hours
1 - explanation of detox eg. toxins
dissolve / leach / diffuse out / water soluble.

Question 17 (3 marks)

Consider the following balanced chemical equation:



- (a) Does the reaction in the equation above proceed spontaneously? Justify your answer.

1 - yes, E° is +ve (0.01V).
OR 1 - yes Sn is more reactive than Pb

1

- (b) Identify the spectator ion in the equation in part (a).

NO_3^- * did not penalise $(\text{NO}_3)_2^-$?

1

- (c) Identify the reducing agent (reductant), for the equation in part (a).

$\text{Sn}(s)$

1

Question 18 (6 marks)**Marks**

A student conducts an investigation to determine the voltage of a galvanic cell made from half-cells containing magnesium and silver, and their respective ions.

- (a) Draw a labelled diagram of the experimental set-up. Indicate on your diagram the direction of electron flow and the direction of cation movement.

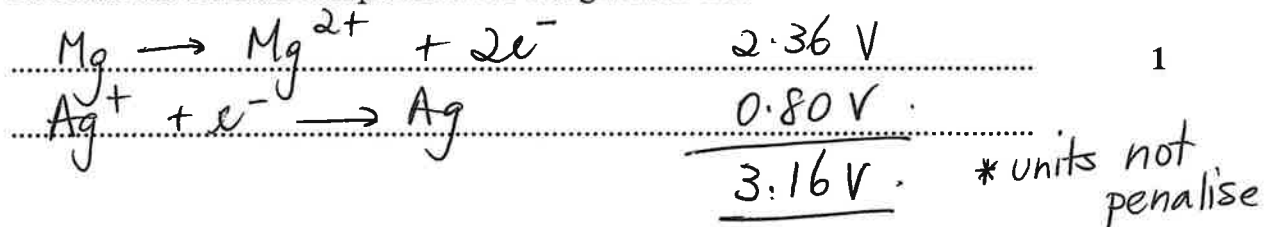
4

- 1 - 2x $\frac{1}{2}$ cells + salt bridge
- 1 - labels (solutions / electrodes)
- 1 - e^- flow (from $Mg(s) \rightarrow Ag(s)$ via external wire)
- 1 - cation flow (across salt bridge from $Mg \rightarrow Ag$ cell)

* ignored lack of $\odot$ or $\otimes$
 * 1 mark given if e^- and cations are same, but in wrong, direction.

- * only 2 max if Mg electrode in Ag^+ / Ag in Mg^{2+} .
 * only 2 max if battery added.

- (b) Calculate the standard cell potential for this galvanic cell.

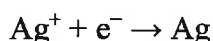


- (c) The experimental voltage obtained from galvanic cells is often different to that indicated from standard reduction potentials. Suggest a reason why this may be.

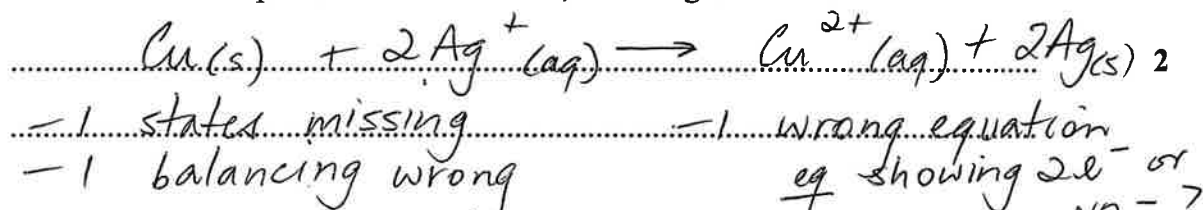
Conditions move away from std quickly. 1
 - any valid reason paid
 * did not pay human error; faulty equipment
 incomplete reaction

Question 19 (5 marks)**Marks**

In a metal displacement reaction following half reactions were observed to occur:



- (a) Write a net ionic equation for this reaction, including all states.



- (b) A strip of copper was placed into a solution containing silver nitrate. After the reaction was complete the silver was removed from the surface of the copper strip and the copper strip was removed. The resulting suspension, containing solid silver, was filtered.

copper strip before reaction	2.298 g
copper strip after reaction	2.216 g
filter paper before filtration	1.662 g
filter paper after filtration	2.002 g

- (i) Calculate the chemical amount (in mol) of silver produced and of copper consumed.

(1) $n(\text{Ag}) = 0.340 / 107.9 = 3.15 \times 10^{-3} \text{ mol}$ 2

(1) $n(\text{Cu}) = 0.082 / 63.55 = 1.29 \times 10^{-3} \text{ mol}$

- (ii) Calculate the experimental mole ratio of silver to copper and account for any deviation from ideal results.

$n(\text{Ag})/n(\text{Cu}) = 2.44:1$ of ideal of 2:1 * not marked 1

(1) As there is too much silver, the sample must not be dry

* did not pay impurities without clarification nor any answer meaning too little silver.

SECTION II: Attempt ALL Questions

Name:

Class:

Answer the questions in the spaces provided.

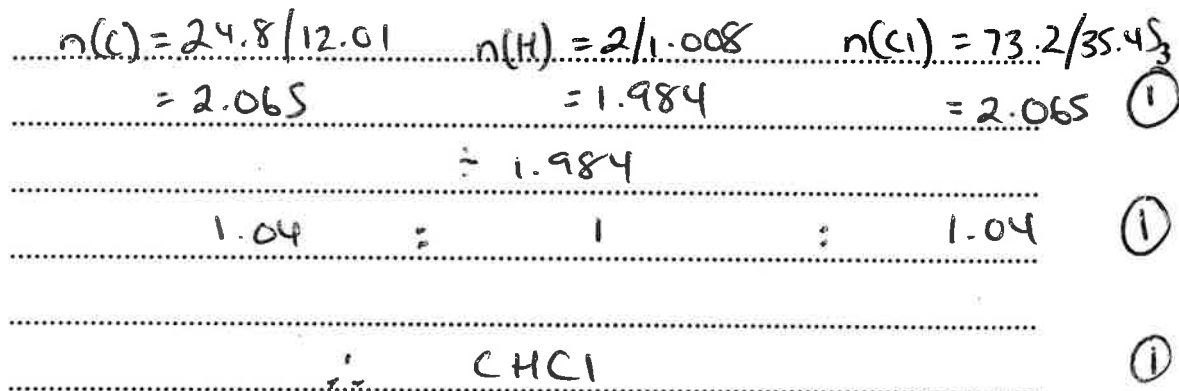
Show all relevant working in questions involving calculations.

Question 20 (4 marks)

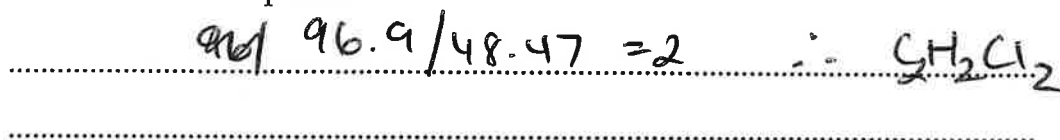
Marks

A compound was found to contain 24.8% carbon, 2.0% hydrogen and 73.2% chlorine.

- (a) Determine the empirical formula of this compound.



- (b) The molar mass of this compound is 96.9 g mol^{-1} . Determine the molecular formula of this compound.



* E.C = carry on error.

Question 21 (2 marks)

Chromium has four stable isotopes, whose relative isotopic masses and percentage abundances are shown in the table below.

Isotope	Isotopic mass (u)	Percentage abundance (%)
^{50}Cr	49.946	4.3450
^{52}Cr	51.941	83.789
^{53}Cr	52.941	9.5012
^{54}Cr	53.939	2.3657

Calculate the relative atomic mass of chromium.

$$\frac{49.946 \times 4.345 + 51.941 \times 83.789 + 52.941 \times 9.5012 + 53.939 \times 2.3657}{100} = 51.997 \quad 2$$

② - correct answer with significant figures.

① - Answer given with incorrect significant figures. Working must be shown for this mark.

- * 14 boys lost marks for calculation errors due to:
- incorrect rounding, particularly intermediate rounding
 - incorrect transcription of values
 - simple calculation errors despite correct working.

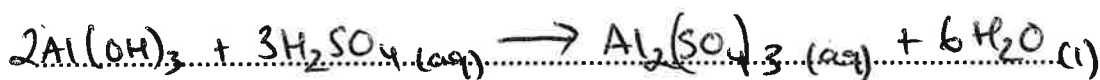
CHECK YOUR WORKING

Many more lost marks for incorrect significant figures.

Question 22 (5 marks)**Marks**

In an experiment, 2.369 g of aluminium hydroxide reacts with 60.0 mL of 0.714 M sulfuric acid to produce a salt solution and water.

- (a) Write a balanced chemical equation for this reaction.



1

* States not marked

- (b) Calculate the chemical amount (in mol) of the salt produced.

$$n \text{Al}(\text{OH})_3 = \frac{2.369}{78.004} = 0.03037 \text{ mol} \quad \text{AND} \quad n \text{H}_2\text{SO}_4 = 0.06 \times 0.714 = 0.04284 \quad \textcircled{1}$$

4

$$\text{Al}(\text{OH})_3 \text{ requires } 0.03037 \times \frac{3}{2} = 0.04556 \text{ mol H}_2\text{SO}_4$$

$\therefore \text{H}_2\text{SO}_4$ limiting. $\textcircled{1}$

$$n \text{Al}_2(\text{SO}_4)_3 = \left[\frac{1}{3}\right] \textcircled{1} \times 0.04284 = 0.0143 \text{ mol} \quad \textcircled{1}$$

Question 23 (4 marks)

A 33.34 g sample of a diatomic gas was stored in a 20.0 L vessel at 150 kPa and 30 °C. What is the identity of this gas?

$$n = \frac{PV}{RT}$$

4

$$= \frac{150 \times 20}{8.314 \times (30 + 273.15)} \quad \textcircled{1}$$

$$= 1.1902 \text{ mol} \quad \textcircled{1}$$

$$M(X_2) = \frac{m}{n} = \frac{33.34}{1.1902}$$

$$= 28.01 \text{ g/mol} \quad \textcircled{1}$$

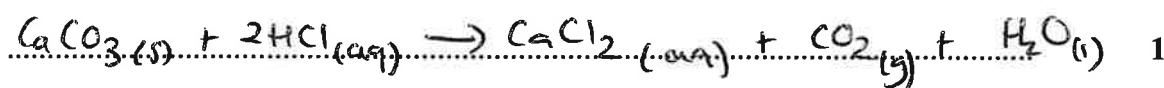
$$M(X) = 28.01/2 = 14.01 \quad \therefore \text{nitrogen} \quad \textcircled{1}$$

Question 24 (5 marks)

Marks

Antacids contain calcium carbonate and are used to neutralise stomach acid. The mass of one antacid tablet was found to be 1.30 g. To determine the amount of calcium carbonate present, this tablet was crushed into a fine powder and excess hydrochloric acid was added to it. After the reaction was complete, there was a decrease in mass of 0.220 g.

- (a) Write a balanced equation for the reaction between calcium carbonate and hydrochloric acid.



* States not marked

- (b) Calculate the mass (in g) of calcium carbonate present in the antacid tablet.

$$n(\text{CO}_2) = 0.220 / 44.01 = 0.004999 \text{ mol} \quad (1) \quad 2$$

$$n(\text{CaCO}_3) = 0.004999 \text{ mol} \quad (1:1)$$

$$m(\text{CaCO}_3) = n \times 100.09 = 0.500 \text{ g} \quad (1)$$

- (c) Calculate the percentage by mass of calcium carbonate in the antacid tablet.

$$0.5 / 1.3 = 38.5\% \quad 1$$

- (d) At 100 kPa and 296 K, what volume of gas would have been produced?

$$V = \frac{nRT}{P} \quad 1$$

$$= \frac{(0.004999 \times 8.314 \times 296)}{100} \div 100 = 0.123 \text{ L}$$

Note: the conditions for 24.79 L is 25°C & 298.15 K which are not the conditions above.

SECTION II: Attempt ALL Questions

Name:

Class:

Answer the questions in the spaces provided.
Show all relevant working in questions involving calculations.

Question 25 (4 marks)

Marks

	silicon tetrachloride	silicon dioxide
Melting point (°C)	-68	1713

Explain the melting points of silicon tetrachloride and silicon dioxide in terms of their structure and bonding.

- ① Identify structure & bonding in SiCl_4
e.g. discrete molecules with 4 Cl atoms covalently bonded to one Si atom.
- ① Identify structure & bonding in SiO_2
e.g. covalent network of Si and O atoms covalently bonded in a 3D array/lattice
- ① Identify what is being broken upon melting
e.g. intermolecular forces between SiCl_4 molecules vs covalent bonds between Si and O atoms.
- ① Relate strengths/energies to melting point data
e.g. covalent bonds require more energy to break than intermolecular forces, hence SiO_2 has a much higher melting point than SiCl_4 .

4

Question 26 (6 marks)**Marks**

5.000 g of silver nitrate is dissolved in 100.0 g of water, initially at 25.0 °C.

- (a) If the enthalpy of solution ($\Delta_{\text{sol}}H$) for silver nitrate is +22.6 kJ mol⁻¹, what is the final temperature of the solution?

① $n(\text{AgNO}_3) = \frac{5.000 \text{ g}}{107.9 + 14.01 + 3 \times 16.00 \text{ g mol}^{-1}} = \frac{5.000 \text{ g}}{169.91 \text{ g mol}^{-1}} = 0.02943 \text{ mol}$ 4

① $q = 0.02943 \text{ mol} \times 22.6 \text{ kJ mol}^{-1} = 0.665 \text{ kJ} = 665 \text{ J}$

① $\Delta T = \frac{-q_{\text{sys}}}{mc} = \frac{665 \text{ J}}{100 \times 4.18} = -1.59 \text{ °C}$

① $\therefore T_{\text{final}} = 25.0 - 1.59 = 23.4 \text{ °C}$

signs not marked for q or ΔT , just direction of final temperature (i.e. subtract off from 25.0).

- (b) Given that silver nitrate is soluble in water at room temperature, what must be the sign of the entropy of solution ($\Delta_{\text{sol}}S$) for silver nitrate? Explain your answer.

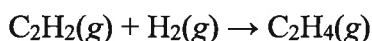
① { Given that silver nitrate dissolves in water, $\Delta_{\text{sol}}H$ must be negative. 2

① { Since $\Delta_{\text{sol}}H$ is positive, the only way $\Delta_{\text{sol}}G$ can be negative is if $-T\Delta S$ is negative, i.e. if $\Delta_{\text{sol}}S$ is positive.

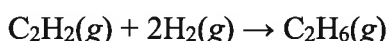
If boys argued that $\text{AgNO}_3(\text{s}) \rightarrow \text{Ag}^+(\text{aq}) + \text{NO}_3^-(\text{aq})$ must have a positive $\Delta_{\text{sol}}S \rightarrow$ ① max. This is not actually true in general for all ionic compounds.

Question 27 (2 marks)**Marks**

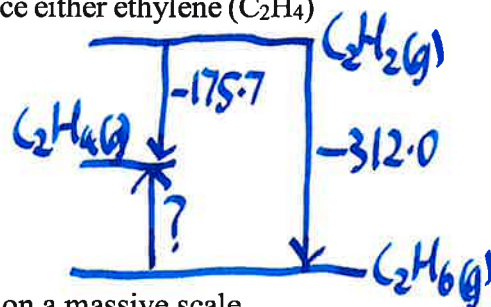
The reaction of acetylene (C_2H_2) with hydrogen (H_2) can produce either ethylene (C_2H_4) or ethane (C_2H_6):



$$\Delta_r H^\ominus = -175.7 \text{ kJ mol}^{-1}$$

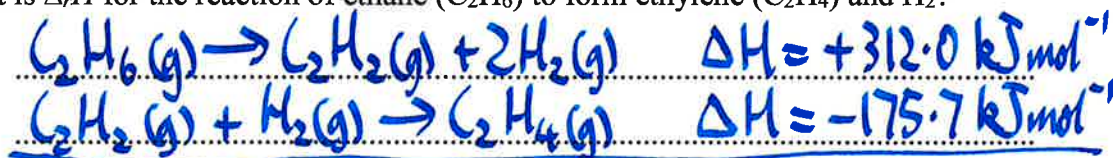


$$\Delta_r H^\ominus = -312.0 \text{ kJ mol}^{-1}$$

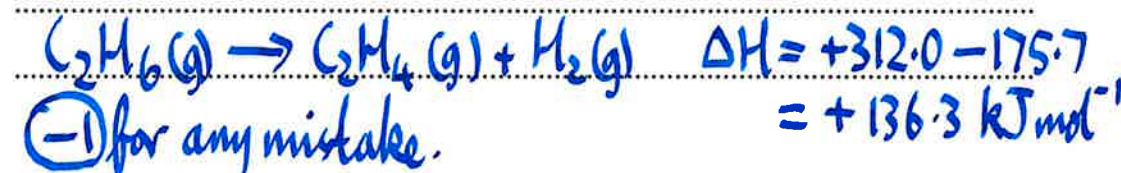


Industrially, ethylene and hydrogen are produced from ethane on a massive scale.

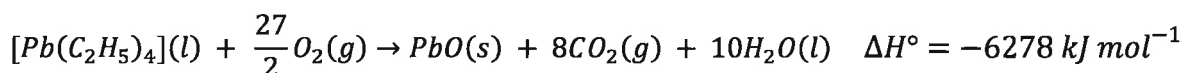
What is $\Delta_r H$ for the reaction of ethane (C_2H_6) to form ethylene (C_2H_4) and H_2 ?



2

**Question 28** (3 marks)

Tetraethyllead $[\text{Pb}(\text{C}_2\text{H}_5)_4]$ was formally used as a fuel additive, but its use in automotive petrol has been banned in Australia since 2002. It combusts as follows:



	standard enthalpy of formation (kJ mol^{-1})
$\text{CO}_2(\text{g})$	-393.5
$\text{H}_2\text{O}(\text{l})$	-285.8
$[\text{Pb}(\text{C}_2\text{H}_5)_4](\text{l})$	+53.0

Given the information in the table above, calculate the standard enthalpy of formation of $\text{PbO}(\text{s})$.

$$\begin{aligned} \Delta H^\ominus &= -6278 \text{ kJ mol}^{-1} = \Delta_f H^\ominus(\text{PbO}) + 8 \times (-393.5) + 10 \times (-285.8) - 53.0 \\ \therefore \Delta_f H^\ominus(\text{PbO}) &= -6278 + 8 \times 393.5 + 10 \times 285.8 + 53.0 \\ &= -219 \text{ kJ mol}^{-1} \end{aligned}$$

3

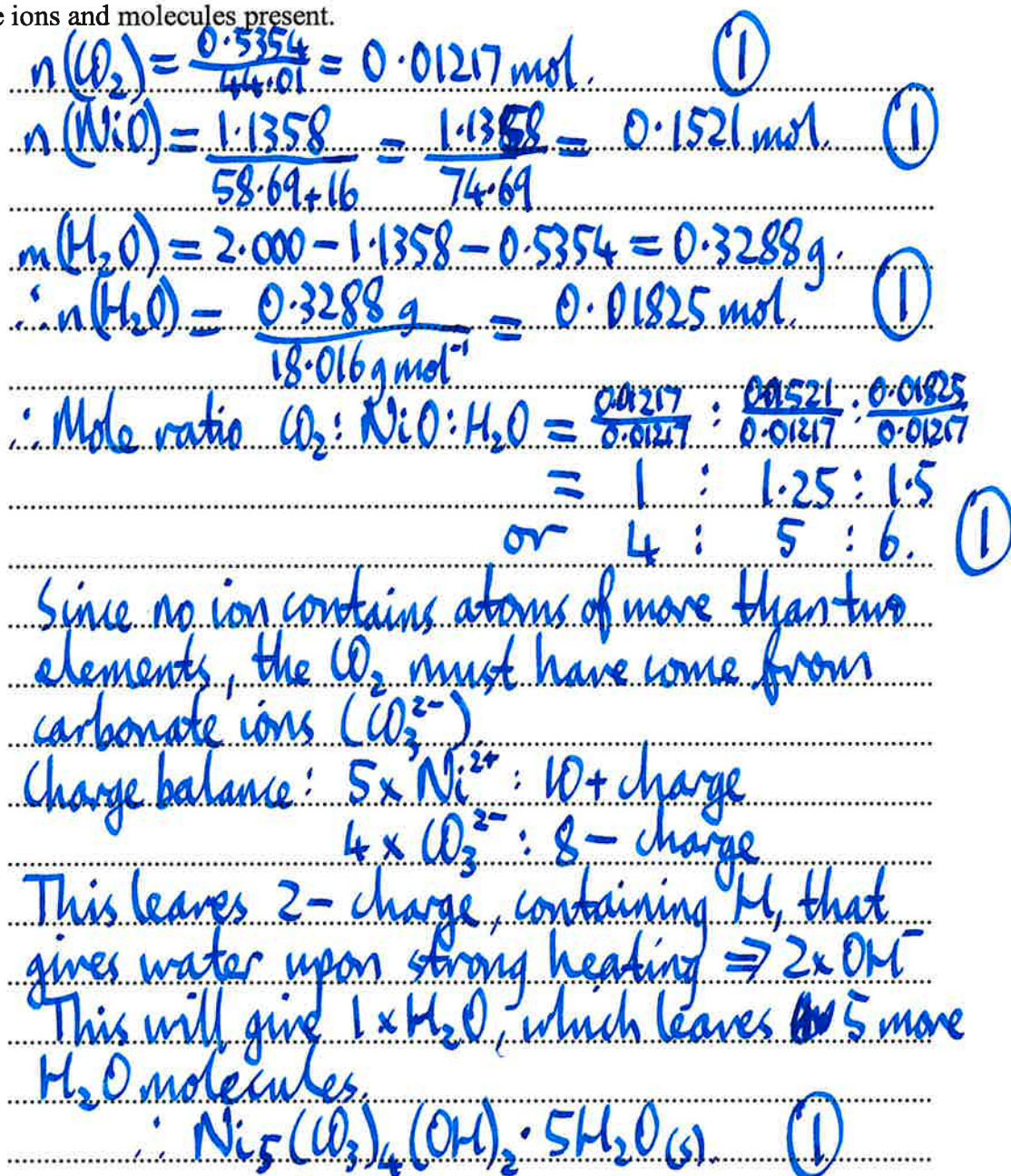
⊖ per error

Question 29 (5 marks)**Marks**

Widgiemoolthalite is a rare blue-green mineral first identified in Widgiemooltha, Western Australia, in 1992. It contains three different ions, none of which contains atoms of more than two elements.

When 2.000 g of widgiemoolthalite reacts with excess hydrochloric acid it dissolves completely and releases 0.5354 g of carbon dioxide. Strongly heating 2.000 g of widgiemoolthalite produces water vapour, carbon dioxide and 1.1358 g of nickel(II) oxide as the only products.

Calculate the empirical formula for widgiemoolthalite, expressing your answer in terms of the ions and molecules present.

**END OF EXAMINATION**